

GTIIT Spring 2025 Calculus 1M

Workshop 3

March 20, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint, a big clue, or the full solution— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

Exercises on Basic Properties of Limits

1. Use the graph of the function f to state the value of each limit. If it does not exist, explain why.

$$\lim_{x \rightarrow 0^+} f(x), \quad \lim_{x \rightarrow 0^-} f(x), \quad \lim_{x \rightarrow 0} f(x)$$

(a) $f(x) = \frac{1}{1 + \exp(\frac{1}{x})}$

(b) $f(x) = \frac{x^2 + x}{\sqrt{x^3 + x^2}}$

2. Determine the limits

(a) $\lim_{x \rightarrow 5^-} \frac{x+1}{x-5}$

(b) $\lim_{x \rightarrow 5^+} \frac{x+1}{x-5}$

(c) $\lim_{x \rightarrow 3} \left(-\sqrt[5]{x} + \frac{e^x}{1 + \ln(x)} + \sin(x) \cos(x) \right)$

(d) $\lim_{x \rightarrow 1} \left(\frac{x^2 + 2x}{x^2 - 2x + 1} \right)$

(e) $\lim_{x \rightarrow 2} \frac{2-x}{\sqrt{x+2}-2}$

(f) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

3. Find the vertical asymptotes of the function:

$$y = \frac{x^2 + 1}{3x - 2x^2}$$

Confirm your answer by graphing the function

4. Application of the Squeeze Theorem:

Define the function

$$f(x) = x^2 \cos\left(\frac{1}{x}\right) \sin\left(\frac{1}{x}\right) \quad \text{for } x \neq 0,$$

and set $f(0) = 0$. Use the squeeze theorem to prove that

$$\lim_{x \rightarrow 0} f(x) = 0.$$

Hint: Find appropriate functions $g(x)$ and $h(x)$ that bound $f(x)$ from below and above, respectively.

5. Absolute Value and Limit Existence:

Construct a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that the limit $\lim_{x \rightarrow 0} f(x)$ does not exist, but the limit $\lim_{x \rightarrow 0} |f(x)|$ exists. Provide a rigorous justification for your example.

6. Lateral Limits, Absolute Value, and Overall Limit:

Consider the function

$$f(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

- (a) Determine $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$.
- (b) Explain why the overall limit $\lim_{x \rightarrow 0} f(x)$ does not exist, and compute $\lim_{x \rightarrow 0} |f(x)|$.

Remark: The function $f(x)$ is also called the sign or signum function and is denoted $\operatorname{sgn} x$

Lecture 6

1. Prove the statement using ε and δ definition of a limit and illustrate it with a graph:

- (a) $\lim_{x \rightarrow 2} 2x - 2 = 2$
- (b) $\lim_{x \rightarrow 4} -7 + \frac{1}{2}x = -5$

2. Prove the statement using ε and δ definition:

- (a) $\lim_{x \rightarrow 3} x^2 = 9$
- (b) $\lim_{x \rightarrow 789} c = c$, where c is some constant.

3. Suppose that $\lim_{x \rightarrow a} f(x) = +\infty$ and $\lim_{x \rightarrow a} g(x) = c$, where c is a real number. Prove each statement.

- (a) $\lim_{x \rightarrow a} (f(x) + g(x)) = +\infty$,
- (b) $\lim_{x \rightarrow a} (f(x)g(x)) = +\infty$ if $c > 0$,
- (c) $\lim_{x \rightarrow a} (f(x)g(x)) = -\infty$ if $c < 0$.
- (d) *Hard question:* If the function $f(x)$ is defined by

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is rational,} \\ 1 & \text{if } x \text{ is irrational.} \end{cases}$$

prove that $\lim_{x \rightarrow 0}$ does not exist.